RAUL VALLE

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Education

University of Florida | PhD, Electrical and Computer Engineering | Expected May 2030
University of Florida | BS in Computer Engineering, cum laude | May 2025

Selected coursework: Modern Analysis; Probability and Potential Theory; Deep Learning; Machine Learning for Time Series; Applied Machine Learning Systems.

Research experience

SmartDATA Lab, University of Florida
Graduate Researcher | Fall 2025-present | Advisor: Joel B. Harley

- Developing learned surrogate models for computational fluid dynamics, with current documented experiments on instantaneous particle-flow field prediction.
- Designed hyperparameter and sensitivity studies using Optuna, Latin hypercube sampling, fixed validation controls and importance/interaction visualizations.

Computational NeuroEngineering Lab, University of Florida
Machine Learning and Model Evaluation | February 2024-present

- Diagnosed collapse-to-zero and state-divergence failures in a coupled EEG estimation and parameter-learning pipeline; added trajectory, parameter and reconstruction diagnostics and tested normalization and regularization interventions.
- Built diagnostics for low-SNR zebrafish imaging using label auditing, whitening and ICA experiments, leakage checks, and controlled comparisons of data-quality and model effects.
- Designed reproducible infrastructure for model sweeps, ablations and sensitivity studies using PyTorch, CuPy and RAPIDS.

Selected ongoing project

Kernel Adaptive Memory | Investigating persistent memory for sequence models through synthetic tasks, transformer baselines, and learned-versus-fixed support and routing ablations. Studying online adaptation with a normalized-LMS readout under a Mackey-Glass dynamics shift.

Preprint

M. K. Maldaner, R. Valle, J. Kim, et al. Plato’s Cave: A Human-Centered Research Verification System. arXiv:2603.23526, 2026. Coauthor.

Teaching and leadership

IEEE Signal Processing Society at UF | Chapter President, 2026-27
Organized workshops, journal clubs and Jupyter curricula; mentored multidisciplinary student research teams in signal processing and machine learning.
UF ECE | Undergraduate Peer Instructor | August 2023-December 2024
Supported machine-learning and DSP instruction through reproducible labs and technical tooling.

Technical skills

Python, C/C++, MATLAB, PyTorch, CUDA, CuPy, RAPIDS, Linux, Git, Jupyter, HiPerGator/HPC; experimental design, ablations, representation learning, kernel methods, state-space models, calibration and uncertainty, hyperparameter optimization.